

CNC Milling: G-Code Programming & Cycle-Time Analysis

Programming, simulating, and machining an aluminum part and a tension specimen on a CNC mill

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Course	ME 215-102, NJIT	Instructor	Prof. Suwattananont
Team	Abdullah Masood (lead), Luis Ramirez, Hayley Hernandez, Stacy Aquino, Samuel Kitoye-Rufus, Annemarie Malek	Tools	CNC vertical mill, G-code simulator, end mills (Ø0.25, Ø0.5, Ø0.625 in)

PROJECT SUMMARY

CNC machining produces complex parts with high accuracy and efficiency. In this project, the team learned CNC programming fundamentals, wrote G-code manuscripts and operation sequences for two parts — an aluminum plate with drilled holes and a milled slot (P-34) and a flat tension-test specimen — and calculated spindle speeds, cutting speeds, and machining cycle times. Programs were verified in a simulator and checked by the TA before running on the machine. The work showed how CNC reproduces a design with very little deviation, and why understanding programming, setup, and maintenance matters in practice.

1. Introduction & Objectives

Computer numerical control (CNC) uses software to direct a machine tool’s motion during manufacturing. Its precision and repeatability make it standard across a wide range of industries, especially for parts with complex geometry. The goal of this project was to move beyond a surface-level understanding of CNC and learn how programs, machine setup, and cutting parameters work together.

The objectives were to understand CNC operating principles, learn to write and run G-code, and analyze the resulting data to understand how program choices affect machining time and part quality.

2. Background

CNC is one of the most significant advances in modern machining. Instead of a machinist guiding every motion by hand, the operator enters coded instructions and the machine carries them out automatically. Numerical control was pioneered by John T. Parsons, whose work beginning in 1949 led to the first numerically controlled milling machines. Modern CNC software and hardware allow machines to run programs consistently and precisely while producing parts at high rates, which improves productivity and reduces the need for rework.

3. Procedure

The project was divided into stages. The team first studied the fundamentals of CNC programming, including program structure, canned cycles, part design, and numbered block sequences. Designs were then reviewed and approved before any machining. After a briefing on machine hazards and safe operation, the team learned to set up part zero, use the control software, and apply spindle speed, feed, milling, and drilling settings, along with techniques for cutting and handling finished pieces.

For each part, the team wrote an operation sequence and a G-code manuscript, determined spindle speeds and cutting velocities, and calculated cycle time. Each program was run in a simulator to verify the tool path before it was approved.

4. Part 1 — Aluminum Plate P-34

Part P-34 is a $4.0 \times 2.5 \times 0.5$ in aluminum plate with three $\varnothing 0.250$ in holes drilled 0.355 in deep, a 0.200 in deep slot with R0.125 ends, and a fifth hole near the upper right corner. All features were cut with a single tool.

Table 1. G-code manuscript for P-34 (inches, absolute coordinates)

Block	Code	X	Y	Z	F	Notes
N5	G00 G90	-1.000	-1.000	0.050		S1500, T1, M03 — start spindle
N10	G00	0.375	0.375	0.050		Rapid over hole 1
N15	G01			-0.355	1.5	Mill hole 1
N20	G00			0.050		Retract
N25	G00	0.875				Rapid to slot start
N30	G01			-0.200	1.5	Plunge slot
N35	G01	1.500			1.5	Mill slot
N40	G00			0.050		Retract
N45	G00	2.000				Rapid over hole 4
N50	G01			-0.355	1.5	Mill hole 4
N55	G00			0.050		Retract
N60	G00	3.345	1.644			Rapid over hole 5
N65	G01			-0.355	1.5	Mill hole 5
N70	G00			0.050		Retract
N75	G00	-1.000	-1.000	0.050		Return to start; M30 end

Table 2. P-34 cycle time (Ø0.25 in tool)

Operation	Spindle (RPM)	Cutting speed (ft/min)	Feed (in/rev)	Feed (in/min)	Travel (in)	Time (min)
Mill hole 1	1,500	98.2	0.001	1.5	0.405	0.27
Mill hole 2	1,500	98.2	0.001	1.5	0.250	0.17
Mill slot 2–3	1,500	98.2	0.001	1.5	0.625	0.42
Mill hole 4	1,500	98.2	0.001	1.5	0.405	0.27
Mill hole 5	1,500	98.2	0.001	1.5	0.405	0.27
Total						1.40

5. Part 2 — Tension Specimen

The second program machined a flat tension-test specimen: a 6.00 in blank with a reduced 0.500 in gauge section. The program uses two end mills (Ø0.5 in and Ø0.625 in), with a tool change (M06) between the first pair of through-cuts and the gauge-section cuts.

Table 3. G-code manuscript for the tension specimen

Block	Code	X	Y	Z	F	Notes
N5	G00 G90	–1.000	–1.000	0.050		S2000, T2, M03
N10	G00	1.0295	–0.300	–0.125		Position beside the part
N15	G01		0.425		1.5	Cut through thickness (1 → 2)
N20	G00	4.9705				Reposition (2 → 3)
N25	G01		–0.300		1.5	Cut through thickness (3 → 4)
N30	G00			0.050		Retract; M05 spindle stop
N35						T1, M06 — tool change
N40	G00	1.343	0.0625			Position at point 5
N45	G01			–0.125	1.5	Plunge (5 → 6)
N50	G01	4.658			1.5	Mill gauge section (6 → 7)
N55	G00			0.050		Retract
N60	G00	–1.000	–1.000			Return to start

Spindle speed and cutting speed are related by $N = 12V / (\pi D)$. At 2,000 RPM this gives $V = 327.25$ ft/min for the Ø0.625 in tool and $V = 261.80$ ft/min for the Ø0.5 in tool. A feed of 0.00075 in/rev at 2,000 RPM gives a feed rate of 1.5 in/min, and each cut time is travel divided by feed rate.

Table 4. Tension specimen cycle time

Tool	Operation	Spindle (RPM)	Cutting speed (ft/min)	Feed (in/rev)	Feed (in/min)	Travel (in)	Time (min)
1	Mill through 1–2	2,000	327.25	0.00075	1.5	0.725	0.483
1	Mill through 3–4	2,000	327.25	0.00075	1.5	0.725	0.483
2	Mill through 5–6	2,000	261.80	0.00075	1.5	0.175	0.117
2	Mill through 6–7	2,000	261.80	0.00075	1.5	3.316	2.211
Total							3.29

Table 5. Machining parameters recorded for the demonstration part (cuts A–C)

Cut	Feed (in/min)	Spindle (RPM)
A	10	2,000
B	25	2,010
C	50	2,025

6. Discussion

One of the most important takeaways was the difference between G-codes. **G00** is rapid positioning: the machine moves quickly from one point to another without cutting, typically to bring the tool from its start point to the beginning of a cut. **G01** is linear interpolation: the tool moves to its destination in a straight line at a controlled feed rate, which minimizes skew and imperfections in the cut. Understanding CNC programming really starts with understanding what each code does.

Before any program was run, the TAs and instructor verified the coordinates, directions, and dimensions. For P-34, the team found that a single tool could produce every hole and the slot, avoiding unnecessary tool changes. For the tension specimen, the team shifted strategy: rather than repeatedly lowering and raising the tool at separate points, the cutter made continuous passes along straight lines through the part thickness, and a second end mill was used to finish the long gauge section. The gauge-section pass (3.316 in at 1.5 in/min) dominates the cycle time, so feed rate is the clearest lever for reducing it.

7. Conclusion

This project provided hands-on experience with CNC programming and a foundation in the fundamentals of the machining workforce. The team learned the key programming elements — positioning, feed rate, tool selection, and coordinate instructions — along with proper tool and fixture handling. Writing, simulating, and running real programs demonstrated how CNC turns a design into a high-quality part and why it is so widely used in engineering.

8. Key Concepts

How do you develop a program for a CNC machine?

Start by defining the part's intended function and assessing the material's size, composition, and properties. A drawing or model of the part is highly useful beforehand. Before programming, determine the tool type, feed rate, spindle speed, depth of cut, and other G-code parameters.

Why is pressurized air used during CNC milling?

In the lab, compressed air cools the tool and clears chips from the cut. This preserves accuracy and surface quality and keeps the work area clean.

What is the difference between NC and CNC?

Numerical control (NC) machines were driven by physical media such as punched tape, with more manual effort and less automation. Computer numerical control uses software and digital G-code with far greater automation, making it more reliable and efficient.